SR.IIT *(MODEL-A)&ALL IIT **Date: 09-01-26**
Time: 3 Hrs **SPL GTM TEST-2** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Two identical photocathodes receive light of wavelengths 3.4×10^{-7} m and 5.4×10^{-7} m respectively. The ratio of stopping potential in two cases is $2:1$. Work function of metal is :

   (A) 0.95 eV (B) 4 eV (C) 5 eV (D) 6 eV

2. An elastic ball of mass m moving with speed u collides elastically with another ball of mass $2m$ at rest. What is the maximum elastic deformation energy during collision?

   (A) $\frac{mu^2}{3}$ (B) $\frac{mu^2}{4}$ (C) $\frac{mu^2}{5}$ (D) $\frac{mu^2}{6}$

3. **Assertion (A)** : A pure semiconductor has negative temperature coefficient of resistance.

   Reason (R) : In a semiconductor on raising the temperature, more charge carriers are released, conductor conductance increases and resistance decreases.

   In the light of the above statements, choose the most appropriate answer from the options given below.

   (A) Both A and R are true and R is the correct explanation of A .

   (B) Both A and R are true but R is not the correct explanation of A .

   (C) A is true but R is false.

   (D) A is false but R is also false.

4. An electron in hydrogen atom and one in a singly ionized helium atom are excited to the state $n = 2$. A photon is emitted when these electrons jump back to the ground state in each case. Then

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(A) Energy of photon is same in both.

(B) Radiations emitted by the helium atom are shifted towards the red as compared to radiations from the hydrogen atom.

(C) Radiations emitted by helium are shifted towards the violet as compared to radiations from the hydrogen atom.

(D) There is no definite relation between the radiations emitted by the two atoms.

5. A geostationary satellite is orbiting around earth at a distance r from the centre of earth. If distance of the satellite from earth increases by 0.5%, time period of satellite:

(A) Remains unchanged

(B) Increases by about 11 minutes

(C) Increases by about an hour

(D) Increases by about 30 minutes

6. A particle oscillates according to equation $x = 10\sin(0.5\pi t)$, x is in meters, t in seconds. The speed of the particle when it is at a distance 4 m from the mean position is:

(A) $2\pi\sqrt{3}\,\frac{m}{s}$ (B) $4\pi\,\frac{m}{s}$

(C) $2\pi\,\frac{m}{s}$ (D) $\sqrt{21}\pi\,\frac{m}{s}$

7. For an ideal gas undergoing a reversible adiabatic compression, (U = internal energy T = Temperature, $\gamma = \frac{C_P}{C_V}$. Choose the correct option: [P - Pressure, V - Volume]

(A) UV^{γ} = constant (B) UP^{γ} = constant

(C) $VU^{\frac{1}{\gamma-1}}$ = constant (D) $TU^{\gamma-1}$ = constant

8. A disc starts rolling down an inclined plane from a height h. The inclined plane makes an angle θ with the horizontal plane. Time taken to reach the foot of the inclined plane is:

(A) $\sqrt{\frac{3h}{g\sin^2\theta}}$ (B) $\sqrt{\frac{3\,h}{g\sin\theta}}$ (C) $\sqrt{\frac{2h}{g\sin^2\theta}}$ (D) $\sqrt{\frac{2\,h}{g\sin\theta}}$

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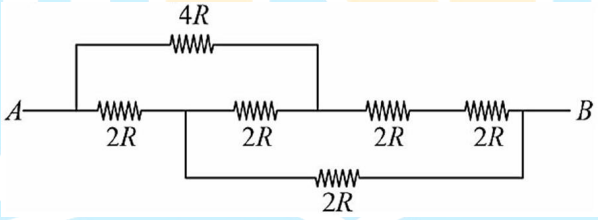
9. In an AC series L-C-R circuit, applied voltage is $V = 100\sqrt{2}\sin\left(\omega t + 45^\circ\right)V$. Given that, $R = 30\Omega$, $X_L = 50\Omega$ and $X_C = 10\Omega$. Now match the following two columns and choose the correct option from codes given below.

List-I		List-II	
(P)	RMS value of current in the circuit	(1)	120 SI unit
(Q)	Average power dissipated in the circuit	(2)	60 SI unit
(R)	RMS value of potential difference across resistance	(3)	2 SI unit
(S)	RMS value of potential difference across inductance	(4)	100 SI unit

(A) $P \to 1, Q \to 2, R \to 3, S \to 4$ (B) $P \to 1, Q \to 4, R \to 3, S \to 2$

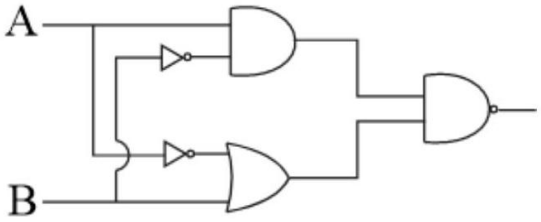
(C) $P \to 3, Q \to 4, R \to 1, S \to 2$ (D) $P \to 3, Q \to 1, R \to 2, S \to 4$

10. Find the equivalent resistance between two ends of the following circuit.



(A) $\frac{3R}{4}$ (B) $\frac{8R}{3}$ (C) $\frac{5R}{3}$ (D) $\frac{6R}{13}$

11. What is the output of the following circuit?

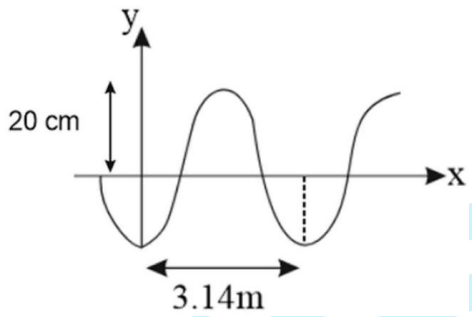


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(A) $\overline{A}+B$ (B) $A\overline{B}+\overline{A}B$

(C) 0 (D) 1

12. A progressive wave of amplitude 20 cm is travelling in the negative x -direction. Its snapshot at $t=0$ is shown in the figure. Frequency of the wave is 5 Hz . The wave equation is given by:



(A) $y=0.2\sin\left(10\pi t-2x+\frac{\pi}{2}\right)m$ (B) $y=0.2\sin\left(10\pi t+2x+\frac{\pi}{2}\right)m$

(C) $y=0.2\sin\left(10\pi t+2x+\frac{3\pi}{2}\right)m$ (D) $y=0.2\sin\left(10\pi t-2x+\frac{3\pi}{2}\right)m$

13. The spectral distribution of the sun has maximum intensity at 4753Å. If the temperature of the sun is 6050 K , what is the temperature of a star for which intensity is maximum at 9506Å ?

(A) 5000 K (B) 1000 K (C) 2050 K (D) 3025 K

14. A film of water is formed between two straight parallel wires of length 10 cm each separated by 0.5 cm . If their separation is increased by 1 mm while still maintaining their parallelism, how much work will have to be done?

(surface tension of water $=7.2\times10^{-2}\ \text{N/m}$)

(A) 7.22×10^{-6} joule (B) 1.44×10^{-5} joule

(C) 2.88×10^{-5} joule (D) 5.76×10^{-5} joule

15. If a bimetallic strip is heated, it will :-

(A) bend towards the metal with lower thermal expansion coefficient.

(B) bend towards the metal with higher thermal expansion coefficient.

(C) twist itself into helix.

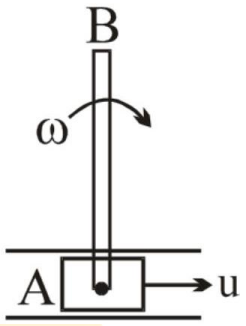
(D) have no bending.

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16. Two equal drops of water are falling through air with a steady velocity v. If the drops coalesced, what will be the new steady velocity?

(A) $2\sqrt{2}v$ (B) 2 v (C) $2^{2/3}v$ (D) $2^{4/3}v$

17. A mechanism consists of a part which is translated with a velocity u and a rod AB of length L and mass M is hinged at A . The rod rotates about axis A with angular velocity ω. The kinetic energy of rod when it is vertical as shown is :



(A) $\frac{1}{2}Mu^2+\frac{1}{6}ML^2\omega^2$ (B) $\frac{1}{2}Mu^2+\frac{1}{6}ML\omega u$

(C) $\frac{1}{2}Mu^2+\frac{1}{6}ML^2\omega^2+\frac{1}{2}ML\omega u$ (D) None of these

18. In telescope, if the powers of an objective and eye lens are +1.25 D and +20 D respectively, then for relaxed vision, the length of tube and magnification will be :-

(A) 85 cm and 25 (B) 85 cm and 16

(C) 21.25 cm and 16 (D) 21.25 cm and 25

19. A power transformer is used to step up an alternating emf of 220 volt to 11 kV to transmit 4.4 kW of power. If the primary coil has 1000 turns, what is the current in the secondary ?

(A) 2.5 A (B) 0.4 A (C) 2 A (D) 0.8 A

20. The SI unit of inductance (Henry) can be written as :-

(i) $\frac{\text{weber}}{\text{Ampere}}$ (ii) $\frac{\text{volt.sec}}{\text{Ampere}}$

(iii) $\frac{\text{Joule}}{(\text{Ampere})^2}$ (iv) ohm-second

(A) (i), (iii) (B) (ii), (iii) (C) (ii), (iv) (D) All

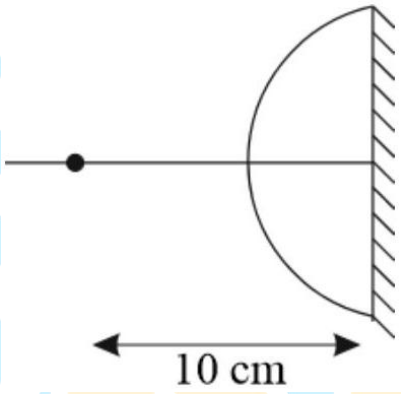
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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

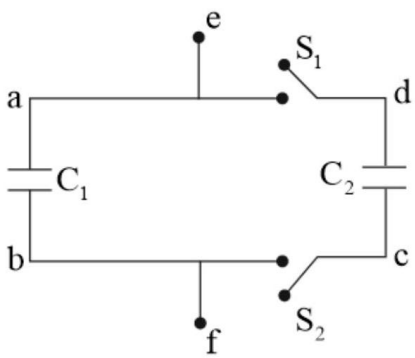
21. A point object is placed in front of a thin planoconvex lens at a distance of 10 cm as shown in the figure. Focal length of the lens in air is 30 cm . The plane surface of the lens is now silvered. The
image created by the system is at x cm to right of the system, where x is ____ .



22. Amplitude of Electric field in light at a distance of 10 m from a LED source is $\sqrt{3}\frac{V}{m}$. What is the power (in watts) of the LED source? Assume the source transmits light uniformly in all directions.

[Take : $c = 3\times10^8 \frac{m}{s}, \frac{1}{4\pi\epsilon_0} = 9\times10^9$ SI units]

23. Capacitor $C_1 = 40\mu$ F and $C_2 = 10\mu$ F are each charged to a potential $V = 50$ V but with opposite polarity, so that points a and c are on the side of the respective positive plates of C_1 and C_2, points b and d are on the side of respective negative plates. Switches S_1 and S_2 are now closed. What is the potential difference (in volts) between e and f ?



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24. What is the electric flux (in Nm^2C^{-1}) through a plane surface of area $10\ m^2$ in plane $y = 2\ m$ kept in an electric field $\vec{E} = \left(-2\hat{i} + 4\hat{j} + 3\hat{k}\right)\frac{N}{C}$?

25. A swimmer can swim with velocity of $12\ km/h$ in still water. Water flowing in a river has velocity $6\ km/h$. The angle (in degree) with respect to the direction of flow of river water that, he should swim in order to reach the point on the other bank just opposite to his starting point is :-

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. " V " mL of $0.1M H_2O_2$ solution is required to completely react with acidified solution of KI . The iodine so liberated required 200 mL of $0.1\ N Na_2S_2O_3$ for titration. The same amount of I_2 is also liberated when " X " millimole KIO_3 is reacted with excess KI . The value of " X " is :

(A) $\frac{3}{10}$ (B) $\frac{10}{3}$ (C) $\frac{5}{3}$ (D) $\frac{3}{5}$

27. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to

(A) The physical state of X_2 at room temperature changes from gas to solid down the group

(B) Increase in HOMO-LUMO gap down the group

(C) Decrease in $n\pi^* - n\sigma^*$ energy gap down the group

(D) Decrease in ionization energy down the group

28. **Assertion :** Phenolphthalein do not show any colour change during the titration of $NaHCO_3$ with HCl .

Reason : In presence of phenolphthalein, HCl do not react with $NaHCO_3$

(A) Assertion is true, Reason is true and Reason is correct explanation for Assertion

(B) Assertion is true, Reason is true and Reason is NOT the correct explanation for Assertion

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(C) Assertion is true, Reason is false

(D) Assertion is false, Reason is true

29. For any sparingly soluble salt $\left[M(NH_3)_4Br_2\right]H_2PO_2$

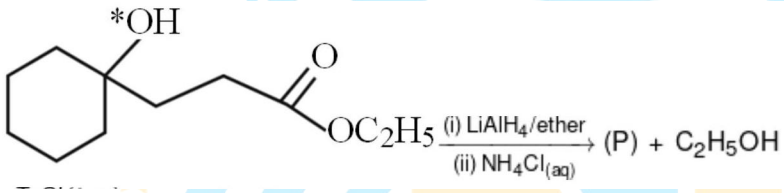
Given :

$\lambda^0_{[M(NH_3)_4Br_2]^+} = 400\ Sm^2mol^{-1}$, $\lambda^0_{H_2PO_2^-} = 100\ Sm^2mol^{-1}$,

Specific resistance of saturated $\left[M(NH_3)_4Br_2\right]H_2PO_2$ solution is $200\Omega\ cm$. If solubility product constant of the above salt is 10^{-x}. The value of 'x' is

(A) 8 (B) 10 (C) 12 (D) 6

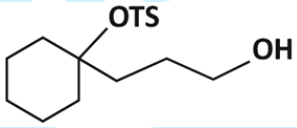
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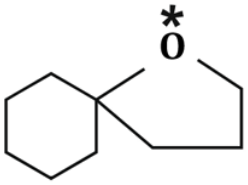


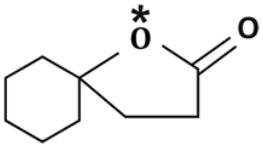
$(P) \xrightarrow[Pyridine]{TsCl\,(1eq)} (Q)\ (major) \xrightarrow[(mild)]{Naoh} (R)$

The correct option is

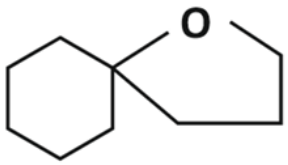
$\left[\overset{*}{O}=\overset{18}{O}\right]$

(A) 'Q' is 

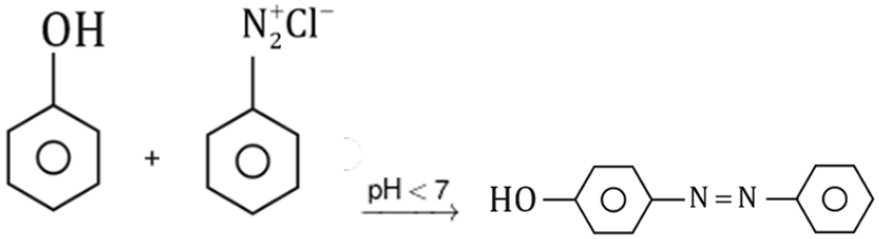
(B) 'R' is 

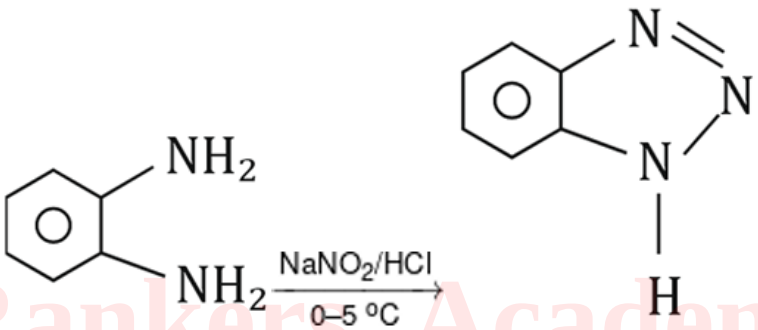
(C) (P) is 

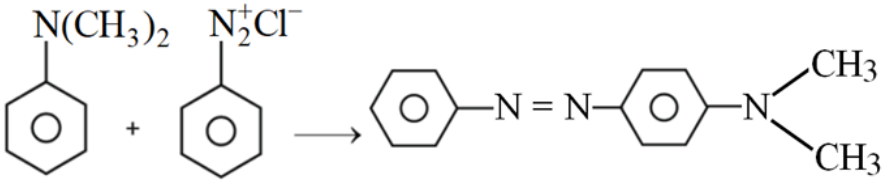
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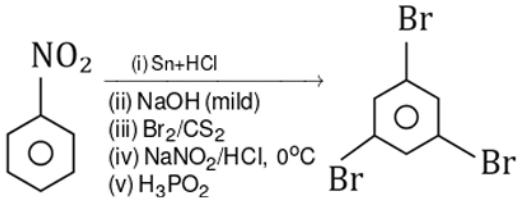
(D) ' R ' is 

31. The Incorrect option is

(A) 

(B) 

(C) 

(D) 

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32.



$\xrightarrow[AlCl_3]{(CH_3)_2CHCl}$ (P) $\xrightarrow[(ii)\ H_3O^+/\Delta]{(i)\ O_2/h\nu}$ (Q) + (R)

Given : (P) is a monosubstituted product and molar mass of ' R ' is greater than (Q)

Which of the following statement is **INCORRECT** ?

(A) ' R ' gives violet colour solution with neutral $FeCl_3$ solution

(B) $\xrightarrow[(ii)H_3O^+(mild)]{(i)NaOH/CHCl_3}$ isomeric mixture of products

Major isomeric product has relatively higher boiling point that other isomeric products formed

(C) 'Q' does not give red precipitate with a mixture of Rochelle's salt/ OH^- and $CuSO_{4(aq)}$

(D) ' R ' is useful in making of resin & Q is used as a solvent.

33. How many moles of NH_3 must be added to 2.0 litre of $0.80M\ AgNO_3$ in order to reduce the Ag^+ concentration to $5\times10^{-8}M$. K_f of $\left[Ag(NH_3)_2\right]^+ = 10^8$

(A) 0.4 (B) 2 (C) 3.52 (D) 4

34. In the hydrogen spectrum, the longest wavelength in the Lyman series as $\mathbf{xA^o}$ and the shortest wavelength in the Balmer series is $\mathbf{yA^o}$. Hence, the longest wavelength of light needed to ionise a hydrogen atom from its ground state should be

(A) $(x+y)A^0$ (B) $(y-x)A^0$

(C) $\frac{xy}{(x+y)}A^0$ (D) $\frac{(x+y)}{xy}A^0$

35. A **1**L reaction vessel which is equipped with a movable piston if filled completely with a 1M aqueous solution of H_2O_2. The H_2O_2 decomposes to $H_2O(l)$ and $O_2(g)$ in a first order process with half life 5 hrs at 300 K . As gas formed, the piston moves up against constant external pressure of 1 atm . What is the net work done by the gas (in cal) from the start of sixth hour till the end of 10 hrs ? ($R = 2\,cal\,mol^{-1}\,K^{-1}$) (Considering gas to be ideal)

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(A) 25 Cal (B) 50 Cal (C) 75 Cal (D) 100 Cal

36. Incorrect option is

(A) In SF_2Cl_2 molecule, axial bond lengths are shorter than equatorial bond lengths

(B) Order of $O\hat{P}X$ bond angle $(X = F, Cl, Br)$: $POF_3 < POCl_3 < POBr_3$

(C) Bond length of $C-O : CO_3^{2-} > H-\overset{\overset{O}{\|}}{C}-\overset{-}{O}$

(D) Order of bond angle: $OCl_2 > NH_3 > H_2O > OF_2$

37.

$$CH_2=C=O \xrightarrow[(2)\,CH_3MgBr\ /\text{ether}\ (\text{excess})]{(1)\,Br_2\,(1eq)/CCl_4} \xrightarrow[(3)\,NH_4Cl_{(aq)}]{}$$

Major product(P)

Which one of the following statement is CORRECT ?

(A) ' P ' decolorises the colour of Br_2/CS_2 solution

' P ' gives white turbidity immediately

(B) with conc. HCl / anhydrous $ZnCl_2$ at room temperature

(C) ' P ' gives colourless gas with $NaHCO_3$ aqueous solution

(D) ' P ' is relatively more acidic than carbolic acid

38. Compound $A, C_5H_{10}O_5$ gives a tetra-acetate with $\left(CH_3-\overset{\overset{O}{\|}}{C}\right)_2O$ and oxidation of A with Br_2-H_2O gives an acid, $C_5H_{10}O_6$. Reduction of A with red P and HI gives iso-pentane, there are two possible structures for compound A which can be distinguished by using

(A) HIO_4 (B) Red $P+HI$

(C) Phenyl hydrazine (D) Tollen's reagent

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39. **Statement-1** : $NH_2 - NH_2$ gives Prussian blue colour during Lassaigne test

Statement-2 : NaCN gives Prussian blue colour of $Fe_4\left[Fe(CN_6)\right]_3$ with $FeSO_4, FeCl_3$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false

(D) Statement-1 is false, statement-2 is true

40. You have a mixture of three amino acids $E\left(P^I = 3.2\right)$, $Y\left(P^I = 5.7\right)$ and $K\left(P^I = 9.7\right)$. Under electrophoresis at $pH = 7.7$, in which direction will each component of the mixture move ? (P^I stands for isoelectric point of amino acid)

(A) E towards anode side ; Y and K towards cathode side

(B) E towards anode side ; Y constant ; K towards cathode side

(C) E towards cathode side ; Y constant ; K towards anode side

(D) E and Y towards anode side; K towards cathode side

41. Which of the following element is not correctly matched with its compound ?

(A) The chloride XCl_3 is oxidizing, but XCl is stable and insoluble in water : **Tl**

(B) The oxide XO_2 has high melting point and is very abundant in nature : Si

(C) X(group 14) forms compounds mainly in +2 oxidation state, through some compounds in the state +4 do exist : Sn

(D) X_4 exists in various allotropic forms and one of the allotrope shows disproportionation reaction with $NaOH : P$

42. Compound ' P ' (orange coloured solid) on heating gives compound ' Q ' (green colour), ' R ' (dry colourless gas) and water. ' R ' on heating with Mg gives a compound ' S ' (white solid) which on hydrolysis gives a colourless pungent smelling gas ' T '. Gas ' T ' gives white fumes with conc. HCl. Which one of the following statement is **INCORRECT** ?

(A) Decomposition of ' P ' is an endothermic reaction

(B) Valency factor of ' P ' in its decomposition reaction is 6

(

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C) Aqueous solution of excess of ' T ', can increase the intensity of colour of $CuSO_4$ aqueous solution

(D) Lattice energy of 'S' is less than that of AlN (Aluminum nitride)

43. Which carbohydrate is used for silvering of mirror ?

(A) Sucrose (B) Cellulose (C) Starch (D) Glucose

44. Number of coordination isomers possible for the given complex $\left[\overset{III}{Co}(en)_3\right]\left[\overset{III}{Cr}(CN)_6\right]$ is :

(including itself)

(A) 6 (B) 4 (C) 2 (D) 3

45. Kipp's waste (obtain from Kipp's arraratus) consists mainly.

(A) conc. H_2SO_4 (B) $Fe_2(SO_4)_3$ (aq.)

(C) $FeSO_4$ (aq.) (D) $CuSO_4$ (aq.)

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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46. The ionization constant of water at $27^\circ C$ is 10^{-14} and at $77^\circ C$ is 2×10^{-14}

$$2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

If magnitude of ' $\Delta H'$ is ' X ' $K\,cal\,mol^{-1}$, then find the value of ' **X** '.

(Assume $\Delta_{ion}H$ is independent of temperature) (Nearest integer)

[Take : $\ln 2 = 0.7$]

47. Co forms dinuclear complex with a sigma bond within two Co atoms. Consider that metal carbonyls follows EAN rule. The complex can be written as $Co_2(CO)_x$. Find the value of x ?

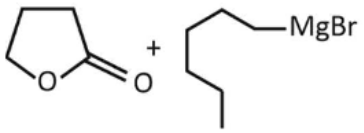
48. The freezing point depression of a 0.10 M solution of formic acid is $0.2046^\circ C$. If the equilibrium constant for the reaction given below at 298 K is 1.1×10^{-x}, then find the value of $'\mathbf{x}'$

$$HCOOH(aq) + H_2O(l) \rightleftharpoons HCOO^-(aq) + H^+(aq)$$

(Given : $K_f(H_2O) = 1.86\ K-kgmol^{-1}$, Molarity = molality)

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49.



1 : 1 $\xrightarrow[\text{(ii) } NH_4Cl_{(aq)}]{\text{(i) either}}$ major product (P)

If number of 3^0 carbon atom(s) per molecule of $'P' = X$, and D.U of $'P' = Y$ then the value of $(X - Y)$ is

50. For 1 mole of an ideal monoatomic gas on moving from one state to another state, the temperature of gas is doubled and its pressure become 4 times. Then entropy change (ΔS) in the process (in J / K) is. (*Nearest integer*)

($\ln 2 = 0.7, R = 8.3$ J / mol – K)

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $f(x) = \begin{vmatrix} x^{2025} & \cos^{2026}x & e^{x^6} \\ \sin^{2025}x & x^{2026} & \sec x \\ \tan^{2025}x & 10 & 20 \end{vmatrix}$ then the value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x)\,dx$ is equal to

(A) 5 (B) 3 (C) 1 (D) 0

52. $S = \left\{ (\alpha, \beta, \gamma) \in Z^3 : \lim_{x \to 0} \left(\frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} \right) = 3 \right\}$.

Then which is true ?

(Note : Z denotes set of integers and $n(X)$ is number of elements in set X }

(A) $n(S) = 2$ (B) $n(S) = 3$

(C) $(-4, 2, 1) \in S$ (D) $(2, -4, 1) \in S$

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53. **Assertion (A)** : There exists two points on the line $\frac{x-1}{1}=\frac{y}{-1}=\frac{z+2}{2}$ which are at a distance of 2 units from the point $(1,2,-4)$.

Reason (R) : Perpendicular distance of point $(1,2,-4)$ from the line $\frac{x-1}{1}=\frac{y}{-1}=\frac{z+2}{2}$ is 1 unit.

(A) **A**, **R** are true and **R** is correct explanation of **A**.

(B) **A**, **R** are true but **R** is not correct explanation of **A**.

(C) **A** is true and **R** is false.

(D) **R** is true and **A** is false.

54. Evaluate the following expression :

$$\frac{1}{25}\sum_{k=1}^{2025}\left[3\sin^{-1}\left(\frac{\sqrt{2}}{\sqrt{8+k}}\right)-\sin^{-1}\left(\frac{16\sqrt{2}+3\sqrt{2}k}{(8+k)^{3/2}}\right)\right]$$

(A) 81π (B) 0 (C) $3\sqrt{3}\pi$ (D) 9π

55. Two finite sets A and B have 'm' and 'n' elements respectively. Number of elements in the power set of A is 112 more than the number of elements in the power set of B. The mean of m,n,t is zero and t is a real number. Standard deviation of m,n,t is $\sqrt{P}$ (where $P\in N$). Then 'P' is

(A) 62 (B) 32 (C) 31 (D) 37

56. Let $a_0=1$ and for $n\geq 1$, define $a_n=3a_{n-1}+1$. Find remainder when $2(a_{2025})+1$ is divided by 5

(A) 2 (B) 1 (C) 3 (D) 4

57. For $I(x)=\int\frac{\sec^2 x-2026}{\sin^{2026}x}dx$, if $I\left(\frac{\pi}{4}\right)=2^{1013}$, then

(A) $3^{1012}\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (B) $3^{1013}\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

(C) $3^{1011}I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (D) $3^{1010}\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

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58. Consider points $A_i(8,i), i \in \{1,2,3,4\}$. Let F be the focus of parabola $y^2 = 4x$. Let P_i be the points on parabola for which perimeter of triangle A_iP_iF minimum. Then the sum of ordinates of P_i for $i \in \{1,2,3,4\}$ is

(A) 4 (B) 8 (C) 10 (D) 20

59. Let $f(x) = \left|1 + e^{|x|} - e^{-x}\right|$ and $g(x) = \left|e^{|x|} - 2\right|$

Match each entry in List-I to the correct entry in List-II

LIST-I		LIST-II	
A)	bounded between $f(x)$ and $g(x)$ is	I)	$e + \frac{1}{e}$
B)	bounded by $y = f(x), x = -1, x = 1$ and the x -axis is	II)	$\ln \frac{16}{e^2}$
C)	bounded by $y = g(x)$ and the x -axis is	III)	$\ln\left(\frac{27}{16}\right)$
D)	bounded by max $\{f(x), g(x)\}$, x -axis, $x = -2$ and y -axis is	IV)	$\ln 27 + e^2 - 7$

The correct option is

(A) (A) - III, (B) - I, (C) - II, (D) - IV

(B) (A) - II, (B) - I, (C) - III, (D) - IV

(C) (A) - III, (B) - IV, (C) - II, (D) - I

(D) (A) - I, (B) - III, (C) - II, (D) – IV

60. $X = \{1,2,3,4,\ldots 2026\}$ and $A \subset X; B \subset X; A \cup B \subset X$ (where $P \subset Q$ denotes that P is proper subset of Q). Then number of ways of selecting unordered pair of sets A and B such that $A \cup B \subset X$.

(A) $\frac{4^{2026} - 3^{2026}}{2}$ (B) $\frac{4^{2026} - 3^{2026} + 2^{2026} - 1}{2}$

(C) $\frac{4^{2026} - 3^{2026} + 2^{2026}}{2}$ (D) None of these

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61. Let $f_n:(-2,1)\to R$ be defined by $f_n(x)=\left|[x+n]\right|+\sqrt{x-[x]}$ and a_n denotes number of points of discontinuity of the function $f_n(x)$. Then $\sum_{r=0}^{2026} a_r$ is equal to (where [.] denotes Greatest Integer Function)

(A) 4 (B) 3 (C) 4052 (D) 2027

62. Let the function $f:(0,2\pi)\to R$ be defined as

$$f(x)=\sin(x)e^{\sin 2x}+\int_0^x e^{\sin 2t}(\sin t-\cos t)dt$$

and n be the number of points of local maximum and local minimum of $f(x)$ respectively, then $(m+n)$ is equal to

(A) 4 (B) 5 (C) 3 (D) 7

63. Let $\vec{a},\vec{b},\vec{c}$ are three unit vectors such that $|\vec{a}+\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=3$ then

(A) $|\vec{a}+2\vec{b}+3\vec{c}|=20$ (B) $|\vec{a}+2\vec{b}+3\vec{c}|<20$

(C) $|\vec{a}+2\vec{b}+3\vec{c}|>20$ (D) $|\vec{a}+\vec{b}-\vec{c}|=1$

64. Let S be the set of all values of λ, for which the shortest distance between the lines $\frac{x-\lambda}{0}=\frac{y-3}{4}=\frac{z+6}{1}$ and $\frac{x+\lambda}{3}=\frac{y}{-4}=\frac{z-6}{0}$ is 13.

Then $8\left|\sum_{\lambda\in S}\lambda\right|$ is equal to

(A) 144 (B) 256 (C) 306 (D) 288

65. Let A and B are two points both lying within a given circle S, and 'P' be a point on circumference of 'S' at which AB subtends the greatest angle.

Statement-1 : If $A\equiv(1,1),B\equiv(1,-1)$ and equation of S is $x^2+y^2=4$ then P will be $(2,0)$

Statement-2 : 'P' will be the point where a circle passing through A and B touches the circle S.

(A) Statement-1 is true, Statement-2 is true

(B) Statement-1 is false, Statement-2 is false

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(C) Statement-1 is true , Statement-2 is false

(D) Statement-1 is false, Statement-2 is true

66. The value of ${}^{12}C_0 - {}^{12}C_2 + {}^{12}C_4 - {}^{12}C_6 + \ldots + {}^{12}C_{12}$ is

(A) 32 (B) -32 (C) 64 (D) -64

67. If x, y, z are positive integers satisfying the system of equations

$xy + yz + zx = 2024$

$xyz + x + y + z = 2025$

then max (x, y, z) is

(A) 504 (B) 224 (C) 674 (D) 606

68. Suppose one of the side of set of equilateral triangle lies along the line : $2025x + 2026y = 2027$ Let m_1, m_2, m_3 be the slope of sides then $m_1 m_2 + m_2 m_3 + m_1 m_3$ equals

(A) -3 for all such triangles (B) -3 for exactly one such triangle

(C) -1 for all such triangles (D) -1 for exactly one such triangle

69. If

$$A^{-1} = \begin{bmatrix} \sin^2\alpha & 0 & 0 \\ 0 & \sin^2\beta & 0 \\ 0 & 0 & \sin^2\gamma \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} \cos^2\alpha & 0 & 0 \\ 0 & \cos^2\beta & 0 \\ 0 & 0 & \cos^2\gamma \end{bmatrix}$$

where $\alpha, \beta, \gamma \in R$ and $C = A^{-5} + B^{-5} + 5A^{-1}B^{-1}\left(A^{-3} + B^{-3}\right) + 10A^{-2}B^{-2}\left(A^{-1} + B^{-1}\right)$

then C is equal to

(A) I (B) 5 I (C) 32 I (D) $2\left(A^{-1} + B^{-1}\right)^2$

70. If $A(z_1), B(z_2), C(z_3), D(z_4)$ are four points in the argand plane that z_1, z_2, z_3, z_4 are the roots of the equation $z^4 - 5z^3 + 18z^2 - 17z + 13 = 0$ and $|z_1| \neq |z_3|$. Let P is the point of intersection of the line $z\left(i + \sqrt{3}\right) + \bar{z}\left(\sqrt{3} - i\right) - 2\sqrt{3} = 0$ with real axis, then value of [PA. PC] is equal to (where [.] denotes greatest integer function)

(A) 2 (B) 3 (C) 4 (D) 5

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $L_i(P_i, m_i)$ denotes line L_i passing through P_i with slope $m_i \forall i \in \{1,2,3\}$. Given $P_1 = (1,1)$, $P_2 = (\sqrt{2}, 2), P_3 = (\sqrt{3}, 3)$ and $m_1 = 2$, $m_2 = 2\sqrt{2}$, $m_3 = 2\sqrt{3}$. Suppose Δ_1 is area of triangle formed by P_1, P_2, P_3. Δ_2 is area of triangle formed by L_1, L_2, L_3. Then $\dfrac{\Delta_1}{\Delta_2}$ is

72. Let

$$S = \left\{ \alpha : \log_2\left(9^{(2\alpha-4)} + 13\right) - \log_2\left(\frac{5}{2} \cdot 3^{(2\alpha-4)} + 1\right) = 2 \right\}.$$

Then the maximum value of β for which the equation

$$x^2 - 2\left(\sum_{\alpha \in S} \alpha\right) x + \sum_{\alpha \in S} (\alpha+1)^2 \beta = 0$$ has real roots, is

73. Let $f : R \to R$ be defined as $f(x) = (2x - 3\pi)^3 + \dfrac{4x}{3} + \cos x$ and $g = f^{-1}$ then the value of $7g'(2\pi) + 3g''(2\pi)$ is (where g' and g'' represent the 1st and 2nd order derivatives of g)

74. Let $f : R - \{1\} \to R - \{1\}$ be a function satisfying the differential equation $2x(y+x)dx - x^2(dx+dy) = (x+y)^2 dx$ with $f(2) = 2$. If area enclosed by $y = f(x)$ and x -axis from $x = 2$ to $x = 3$ is $(a + \ln b)$ where $a, b \in N$, then the value of $(a+b)$ is

75. Let ' L ' be a line passing through $A(1,1,1)$ and intersects lines $x + y = 1, z = 0$ and $x - y = 2, y + z = 3$. If 'θ' be the acute angle between the line L and line $\dfrac{x}{1} = \dfrac{y}{2} = \dfrac{z-1}{3}$ and the value of $\cos\theta$ is $\dfrac{3\sqrt{n}}{\sqrt{14}\sqrt{8}}$, then [n] is (where [.] denotes the greatest integer function)

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